

# 17.7

## Kubernetes Storage

PV, PVC, StorageClass & CSI Drivers

*Java Backend Architecture Interview Series • Tutorial Edition • Easy Diagrams & Examples*

## Why Kubernetes Storage is Different

Containers are stateless by default — when a pod dies, all data inside it is gone. Kubernetes Storage separates the data (PersistentVolume) from the pod, so data survives pod restarts, rescheduling, and even node failures.

■ **Simple Analogy:** A USB drive (PersistentVolume) can be plugged into any laptop (pod). When the laptop is replaced, the USB still has all your files. A PersistentVolumeClaim is your 'request' for a USB drive of a specific size.

## Key Storage Concepts

|                                    |                                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>PersistentVolume (PV)</b>       | Actual storage resource — an EBS volume, NFS share, or local disk. Created by admin or dynamic provisioner. |
| <b>PersistentVolumeClaim (PVC)</b> | A pod's request for storage. Says: 'I need 10Gi of fast SSD storage.'                                       |
| <b>StorageClass</b>                | A template for dynamic provisioning. Defines type (SSD/HDD), cloud provider, reclaim policy.                |
| <b>CSI Driver</b>                  | Container Storage Interface — plugin that connects K8s to cloud storage (AWS EBS, GCP PD, Azure Disk).      |
| <b>Access Modes</b>                | ReadWriteOnce (one pod), ReadOnlyMany (many pods read), ReadWriteMany (many pods read+write).               |

## Diagram — Dynamic Provisioning Flow

| Step           | What Happens                                                           |
|----------------|------------------------------------------------------------------------|
| 1. Pod created | Pod references a PVC (persistentVolumeClaim)                           |
| 2. PVC created | PVC requests: 10Gi, StorageClass=fast-ssd, ReadWriteOnce               |
| 3. Provisioner | StorageClass provisioner (e.g., AWS EBS CSI) creates a real EBS volume |
| 4. PV created  | K8s creates a PV object representing the EBS volume                    |
| 5. PVC binds   | PVC binds to PV — now the pod can mount it                             |
| 6. Pod running | Pod mounts the volume at /data — data persists across restarts         |

## PVC YAML Example

```
# PersistentVolumeClaim
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: postgres-data
spec:
  accessModes: [ReadWriteOnce]
  storageClassName: gp3 # AWS EBS gp3
  resources:
    requests:
      storage: 50Gi

# Mount in Pod
volumes:
- name: data
  persistentVolumeClaim:
    claimName: postgres-data
containers:
- name: postgres
  volumeMounts:
    - mountPath: /var/lib/postgresql/data
      name: data
```

## Storage Classes — SSD vs HDD

| StorageClass   | Cloud | IOPS        | Use Case                              |
|----------------|-------|-------------|---------------------------------------|
| gp3 (EBS)      | AWS   | 3000-16000  | Databases, high IOPS apps             |
| io2 (EBS)      | AWS   | 64000       | Mission-critical DBs (very expensive) |
| standard (GCP) | GCP   | Medium      | General purpose                       |
| pd-ssd (GCP)   | GCP   | High        | Databases on GKE                      |
| azureDisk      | Azure | Medium-High | Azure workloads                       |

■ **Real-World Example:** MongoDB Atlas on Kubernetes uses StatefulSets with volumeClaimTemplates — each MongoDB pod gets its own PVC (e.g., data-mongo-0, data-mongo-1, data-mongo-2). These PVCs are backed by AWS io2 EBS volumes with 16,000 IOPS. If a pod restarts or is rescheduled, it reattaches to its original volume automatically.